

SERGEY CHERNYAKOV

Senior Blockchain / Web3 Engineer · Solidity · DeFi / MEV · Solana & Ethereum

Email: chernyakov.sergey@gmail.com · Phone: +90 535 037 02 32 · Telegram: @imhotepus
LinkedIn: linkedin.com/in/sergey-chernyakov-458506400 · GitHub: github.com/sergeychernyakov
Location: Wrocław, Poland · Remote / EU (open to relocation, hybrid 1–2 days/week OK)

PROFESSIONAL SUMMARY

Senior engineer with **20+ years** of commercial backend experience, with the last 5 years focused on **blockchain infrastructure: Ethereum (Solidity smart contracts, Flashbots, Uniswap V2/V3 + Sushiswap + Pancakeswap + Doorswap arbitrage)** and **Solana (DEX bundlers, sandwich bots, ML-driven memecoin scoring)**. Hands-on with **MEV strategies, flashloan-based arbitrage with zero initial capital, private TX mempools, multi-DEX routing, multicall-based combinatoric search at scale (4,750 combinations/min, 95x perf optimization)**.

Strong **production engineering** foundation underneath: Python (Web3.py), TypeScript (ethers.js, Solana web3.js), MongoDB / PostgreSQL, asyncio pipelines, MEV-protected transaction submission, schema-versioned ML scoring pipelines with **100% backtest-live parity**. Top Rated on Upwork — **18,000+ hours, 71+ contracts**.

Available as **B2B contractor via Deel.com**.

KEY SKILLS

- **Blockchains:** Ethereum (mainnet + L2) · Solana · BNB Chain · Polygon
- **Smart contracts: Solidity** (custom arbitrage contracts) · Web3.py · ethers.js · Hardhat · Foundry · OpenZeppelin
- **DEXs & DeFi:** Uniswap V2 / V3 · Sushiswap · Pancakeswap · Doorswap · Raydium · Serum · 1inch · Balancer (flashloans) · Curve Finance
- **MEV & private mempools: Flashbots Auction** (eth_sendPrivateTransaction) · bundled transactions · sandwich strategies · circular arbitrage · gas-aware tx routing
- **Performance: Multicall3-based combinatoric validation** (95x faster, 4,750 combinations/min) · token database refresh via 1inch API · cron-driven opportunity scanning
- **Tools / SDKs:** Web3.py · ethers.js · Solana web3.js · Anchor framework · Helius API
- **Production engineering:** Python asyncio · TypeScript · MongoDB · PostgreSQL · Docker · Linux · rotating-file logging · retry strategies · gas estimation pre-tx
- **ML for crypto:** XGBoost · feature engineering for token scoring · schema versioning · 5-seed feature-stability protocol · 100% backtest-live decision parity verification
- **Exchanges (CEX integrations):** Binance · Latoken · Cointiger · Azbit · Bitubu — WebSocket + REST APIs

PROFESSIONAL EXPERIENCE

AI Engineering Lead — Sabermine (United Kingdom, Remote)

November 2024 – Present

- Lead AI engineering and automation including AI-driven analytical pipelines used on crypto markets
- Designed and operated Python automation pipelines and integrations
- ML model lifecycle, schema versioning, drift detection

Senior Blockchain / Backend Engineer — Upwork (International Clients, Remote)

June 2012 – Present (Top Rated)

18,000+ hours · 71+ contracts. Crypto / DeFi engagements since 2020:

ML-driven trading systems

- **Solana Pulse Bot** (solana_pulse_bot, archived 2026-05) — Real-time observer bot for Solana memecoin launchpads (**pump.fun, letsbonk**). Two-phase scoring (fast filter @ T+5s + full analysis @ T+90s). **8 XGBoost ML heads:** entry main / entry @T+30 / entry regression / entry-timing (3-class WAIT/BUY/SKIP at 15s checkpoints) / survival hazard / SL+TP+max_hold quantile heads. **82 features per token** (schema v21), time-aware @30/@60/@90 + deltas. **100% backtest-live decision parity** verified on 300+ tokens. Deterministic parallelism (state updates sequential in main loop, scoring parallel on frozen snapshots). Daily 9 honesty checks (shuffled labels, prior drift, economic backtest, calibration check, label noise floor). 5-seed feature-stability protocol for pruning. Stack: Python, asyncpg + psycopg2, Streamlit dashboards, Helius API, WebSockets.

⌘ Ethereum DeFi infrastructure

- **DEX Arbitrage Bot v2** (arbitraflashbot_v2, private) — Advanced arbitrage bot exploiting price differences across **Uniswap V2/V3, Sushiswap, Doorswap, Pancakeswap and others**. Uses **Balancer flashloans** for zero-capital arbitrage (no upfront capital, no risk). **Multicall3-based combination validation — 95× faster** than baseline (4,750+ combinations/min). 1inch API for token DB freshness. MongoDB for valid pairs / opportunities. **Flashbots** for private gas-optimized transactions. Cron orchestration for continuous scanning. **Circular arbitrage** support with mandatory validation requirements.
- **Ether Bundle Bot** (ether_bundle_bot, private) — Node.js / ethers.js Flashbots bundling. Creates deployer / executor / bundle wallets from private keys, calculates required Ether per wallet with dynamic gas, distributes funds, executes bundled transactions (open trading + initial token purchase), optionally collects remaining Ether.
- **Private Transaction Sender** (private_transaction_sender) — Python + Web3.py + **Flashbots Auction API** (eth_sendPrivateTransaction); MEV-protected transaction submission for major DeFi protocols.
- **Sell BABL Bot** (sell_babl_bot) — Uniswap V2 token-sell bot. Continuous monitoring of token balance, gas-fee estimation before tx, swapExactTokensForETH execution with increased priority fees, allowance check + unlimited-allowance setting if needed, rotating-file logger.
- **Arbitrage Bot with Smart Contracts** (arbitrabot, private) — Python orchestrator + custom **Solidity ArbitrageContract.sol**. Cross-exchange arbitrage via deployed smart contract; Binance / Sushiswap / Uniswap API wrappers.

🔗 Solana DeFi

- **Solana Bundle Bot** (sol_bundle_bot, public) — TypeScript bot managing **multiple Solana wallets**. Create / use-existing wallets from .env, top-up wallets, buy / sell / bundle tokens via CLI interactive interface.
- **Solana Bundle Bot v2** (sol_bundle_bot_2, public) — TypeScript advanced bot for **Raydium / Serum DEX** interactions: automated liquidity-pool management, bundled transactions (save fees + improve speed), lookup-table optimization for efficient on-chain ops.
- **Solana Sandwich Bot** (sol_sandwich_bot, public, ★7 — highest-starred repo) — Python sandwich bot.

🔗 CEX integrations

- **Bitubu Exchange Bot** (bitubu_exchange_bot) — Ruby trading bot. WebSocket + REST integration across **Bitubu, Binance, Bitfinex, Latoken, Cointiger, Azbit** exchanges.

🔗 Crypto data tools

- **Dexscreener Bot** (dexscreener) — Telegram bot for dexscreener.
- **Curve Price Fetcher** (curve_price_fetcher) — Curve Finance price fetcher.

Earlier — long-term Senior Backend / Full-stack roles

2004 – 2024

Twenty years of production backend at SaaS, fintech, marketing automation, e-commerce and media companies (ClickFunnels, Fit2you, Lawdingo, Gleam.io, Skykillers, ITC Publishing, RBC-Ukraine, Obozrevatel) — the foundation for the engineering discipline applied to crypto systems (testing, observability, schema versioning, deterministic pipelines, MEV-aware design).

SELECTED OPEN-SOURCE / PORTFOLIO PROJECTS

- **solana_pulse_bot** — Real-time ML scoring (8 XGBoost heads, 82 features, 100% backtest-live parity). github.com/sergeychernyakov/solana_pulse_bot
- **private_transaction_sender** — Flashbots private TX submission. github.com/sergeychernyakov/private_transaction_sender
- **sol_bundle_bot** — Solana multi-wallet CLI manager. github.com/sergeychernyakov/sol_bundle_bot
- **sol_bundle_bot_2** — TypeScript Raydium/Serum bundler. github.com/sergeychernyakov/sol_bundle_bot_2
- **sol_sandwich_bot** (★7) — Solana sandwich bot. github.com/sergeychernyakov/sol_sandwich_bot
- **sell_babl_bot** — Uniswap V2 token-sell bot with gas estimation. github.com/sergeychernyakov/sell_babl_bot
- **bitubu_exchange_bot** — Ruby multi-exchange CEX bot. github.com/sergeychernyakov/bitubu_exchange_bot
- **dexscreener** — Telegram bot for dexscreener.

Private flagship: arbitraflashbot_v2 (DEX arbitrage with Balancer flashloans + Multicall3 95× perf), ether_bundle_bot (Ethereum token launch+bundle via Flashbots), arbitrabot (Python + custom Solidity arbitrage). Available to walk through architecture on call.

EDUCATION

Bachelor's Degree — Don State Technical University, 1999

LANGUAGES

Ukrainian — Native · English — C1 (Fluent professional) · Polish — Learning (A1)

ENGAGEMENT

- **B2B contractor** — invoicing via **Deel.com** (current setup; open to other platforms)
- **Format:** Remote (preferred) · Hybrid OK (1–2 days/week)
- **Location / mobility:** Wrocław, Poland · Open to relocation within EU · Visa sponsorship welcome for non-Poland EU
- **Notice period:** ~1 month